

Freight Shipment Delay Prediction System

Final project presentation

Using machine learning to predict logistics shipment delays



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Introduction

- Global logistics relies on **timely shipment movement** to ensure smooth supply chain operations.
- However, **shipment delays** are frequent in international freight.
- These delays impact **cost, planning, and customer satisfaction**.
- This project focuses on predicting shipment delays using **machine learning**.



! Problem statement

- Shipment delays are frequent in global logistics
- Caused by port congestion, carrier disruption, weather etc., .
- Leads to high costs, poor planning and customer dissatisfaction

🎯 Project Objective

- Build a machine learning model
- Predict on time vs delayed shipments
- Analyze shipment attributes and delay patterns

💡 Solution

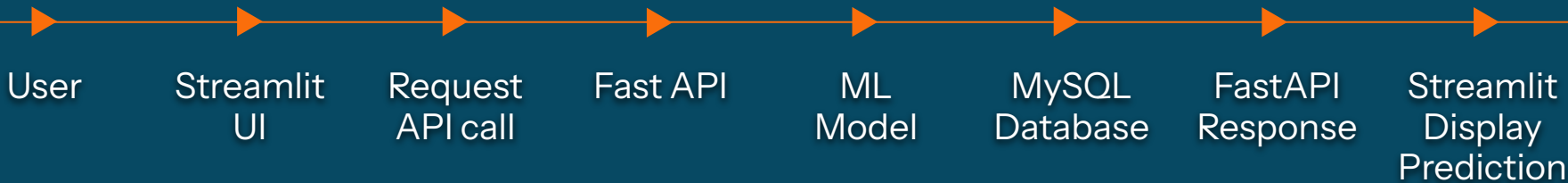
- ML based delay prediction system
- Use shipment input (origin, carrier, shipment date)
- Provides real-time predictions via web application



► Solution overview

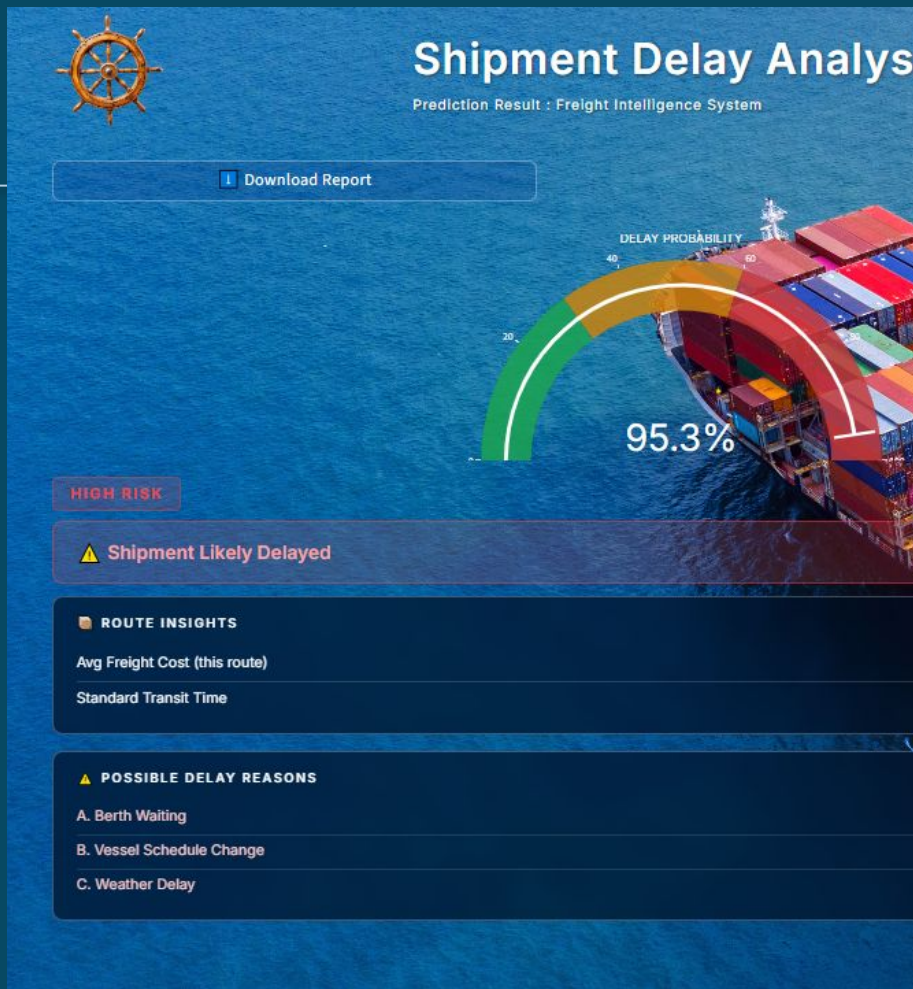
Freight shipment delay prediction system

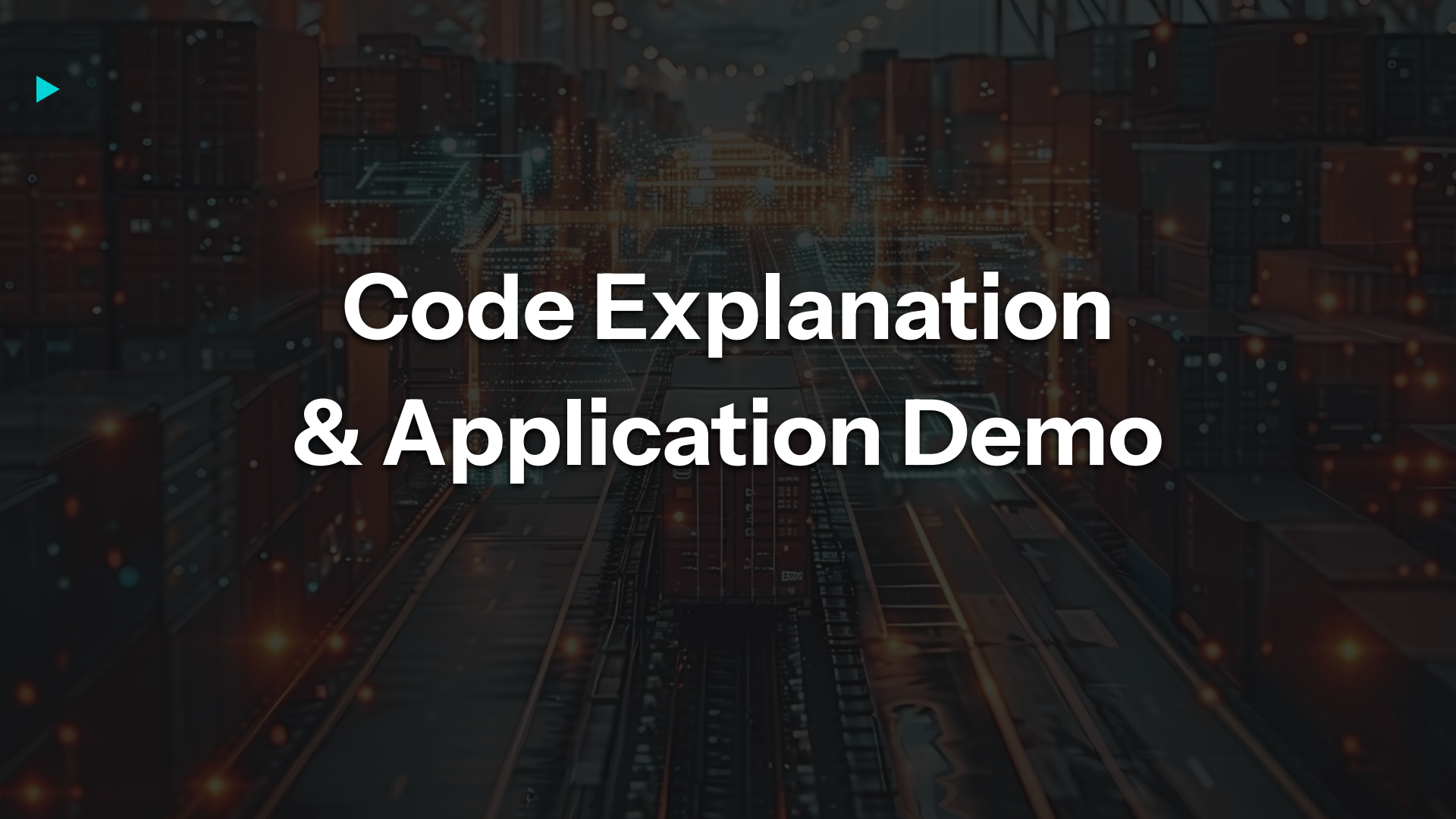
- A machine learning-based system designed to predict shipment delays in advance
- **Takes shipment inputs such as:**
 - Locations (Origin and Destination)
 - Carrier
 - Container type
 - Shipment data
- Processes data using trained ML models
- **Outputs prediction:** On Time or Delayed



Features

- Allows users to input **limited shipment details** to predict delays
- Provides **delay prediction (On Time / Delayed)** using trained ML models
- Displays delay **probability indicator** for better decision-making
- Shows **average transit time** based on shipment data
- Provides **average freight cost insights**
- Enables users to **download results** as a text file
- Offers a **“Learn More”** option to view historical shipment insights
- Includes a **“How to Use”** guide for better user understanding





Code Explanation & Application Demo

Thank you

